

Date	Topics	Homework & Project	Notes
Week 1 Tue, Jan 13	Introduction, Overview of AI		AIMA Ch1
Week 1 Thu, Jan 15	Intelligent Agents	HW1 out: rule-based agent	AIMA Ch2
	Topic 1: Problem Solving		
Week 2 Tue, Jan 20	Problem solving by search: Uninformed search		AIMA 3.1-3.4
Week 2 Thu, Jan 22	Problem solving by search: Informed search		AIMA 3.5, 3.6
Week 3 Tue, Jan 27	Logic & Inference	HW2 out: search-based agent	AIMA Ch7
	Topic 2: Reasoning with Uncertainty		
Week 3 Thu, Jan 29	Rivanna Access, Agentic Framework Tutorial		
Week 4 Tue, Feb 3	Quantifying uncertainty		AIMA Ch13
Week 4 Thu, Feb 5	Probabilistic reasoning: Bayes Nets, Inference by Enumeration	Project: Team and topic registration	AIMA 14.1-14.4
Week 5 Tue, Feb 10	Probabilistic reasoning over time: Markov Process, Inference in Temporal Models	HW3 out - Bayes Net agent	AIMA 15.1-15.3
	Topic 3: Reinforcement Learning		
Week 5 Thu, Feb 12	Reinforcement learning: Markov Decision Process		AIMA 17.1
Week 6 Tue, Feb 17	Reinforcement learning: Policy, Value, Value Iteration		AIMA 17.2
Week 6 Thu, Feb 19	Reinforcement learning: Model-based, Model-free, Q-learning	Project: Agent storyboard due	AIMA Ch21

Week 7 Tue, Feb 24	<i>Midterm Review</i>	HW4 out: RL agent	
Week 7 Thu, Feb 26	Midterm 1		
Week 8 Tue, Mar 3	<i>Spring recess</i>		
Week 8 Thu, Mar 5	<i>Spring recess</i>		
	Topic 4: Learning from examples		
Week 9 Tue, Mar 10	Learning from examples: Naive Bayes, Train/Validation/Test		AIMA 20.1-20.2
Week 9 Thu, Mar 12	Learning from examples: Features, Perceptron	Project: Technical proposal due	AIMA 18.6.3, 18.7.1-18.7.2
Week 10 Tue, Mar 17	Learning from examples: Logistic Regression, Regularization		AIMA 18.4, 18.6.4
Week 10 Thu, Mar 19	Learning from examples: Neural Networks	HW5 out: Neural Network	AIMA 18.7
Week 11 Tue, Mar 24	Learning from examples: Non-parametric models, kNN		AIMA 18.8.1
Week 11 Thu, Mar 26	Rivanna GPU, Neural Network Tutorial		
	Topic 5: AI Applications & Future		
Week 12 Tue, Mar 31	Application: Robotics		
Week 12 Thu, Apr 2	Project working time + milestone feedback	Project: milestone due on Apr 1 for feedback on this date	
Week 13 Tue, Apr 7	Midterm 2		

Week 13 Thu, Apr 9	Project working time + milestone feedback	Project: milestone due on Apr 8 for feedback on this date	
Week 14 Tue, Apr 14	Application/GenAI: Natural language processing, Language generation		
Week 14 Thu, Apr 16	Application/GenAI: Computer vision, Image generation		
Week 15 Tue, Apr 21	GenAI: Generative agents, Issues		
Week 15 Thu, Apr 23	Responsible AI: Explainability, Bias, Fairness	Project: Video due on 4/25 (Sat)	
Week 16 Tue, Apr 28	<i>Course end</i> Project Spotlight		
Week 16 Thu, Apr 30		Project: Final report due	

Generative AI Policy

In this course, we will explore different topics in AI, and of course, how to use generative AI tools is of interest. We also have some opportunities for you to practice with these tools. I encourage you to explore generative AI tools to help your learning, but very importantly, use them **with a critical and responsible view**. That means

- DO NOT simply copy the review questions, homework assignments, and lecture videos to ask AI to generate answers and code to submit without working on them yourself.
- DO NOT use AI to directly generate reports without thinking about the project and working on it yourself.
- DO discuss with AI to clarify or explain the concepts you do not understand well, and if uncertain, ask and discuss on Piazza or during office hours!